



Course Code	CSA0314	Course Title	Data Structures
CO Assessed	CO1	Assessment	Assessment Tool2: Technical presentation
Weightage	35% of CIA	Total Marks	100
CO1 Statement	Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity. (BL2, BL3)		
Student Name	T. Tarun	Reg. No.	1924260001
Section/Batch	Slot C / 2026	Date	03-10-2026

Code of Conduct:

I T. Tarun, certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

T. Tarun

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts	
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established	
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure	
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence	
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand	



SIMATS ENGINEERING
Saveetha Institute of Medical and Technical Sciences
Chennai-602105.



Technical Presentation

UNIVERSITY ATTENDANCE SYSTEM USING QUEUE

CSA0314 - DATA STRUCTURES

Guided by:
Dr. Krishnaraj Mony

Presented By :
T. Tarun(1924260001)



Problem & Requirement

Problem:

1. Attendance is recorded student by student.
2. Students are recorded in sequential order.
3. New attendance records arrive continuously.
4. The system needs efficient insertion of new records.

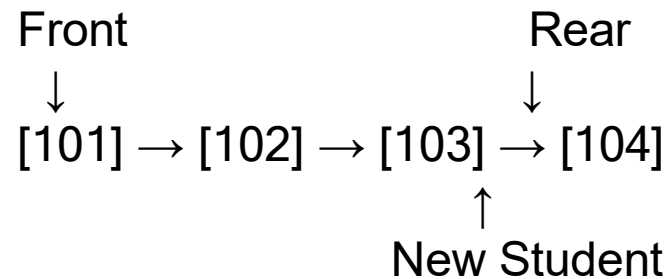
Requirements:

1. Store student attendance.
2. Maintain sequential order.
3. Insert new attendance records.
4. Process records in order.



Suitable Data Structure: Graph

- A Queue is a linear data structure.
- It follows FIFO (First In, First Out).
- Front → First attendance record.
- Rear → Latest attendance record.
- New attendance is inserted at the Rear





Insertion Using Enqueue Operation



Steps:

Select the rear position.

Insert the new attendance record.

Move the rear position.

Continue recording the next student.

Before:

[101] → [102] → [103]

New Student: 104

After:

[101] → [102] → [103] → [104]



Why Queue is Appropriate?

Data structure Suitability

DATA STRUCTURE	SUITABILITY
ARRAY	MODERATE
LINKED LIST	SUITABLE
STACK	NOT SUITABLE
TREE	LIMITED
GRAPH	NOT SUITABLE
QUEUE	HIGHLY SUITABLE



REASON FOR QUEUE

Reason:

- Maintains sequential order.
- New records are inserted at the rear.
- Follows FIFO.
- Provides efficient insertion.
- Enqueue operation $\rightarrow O(1)$

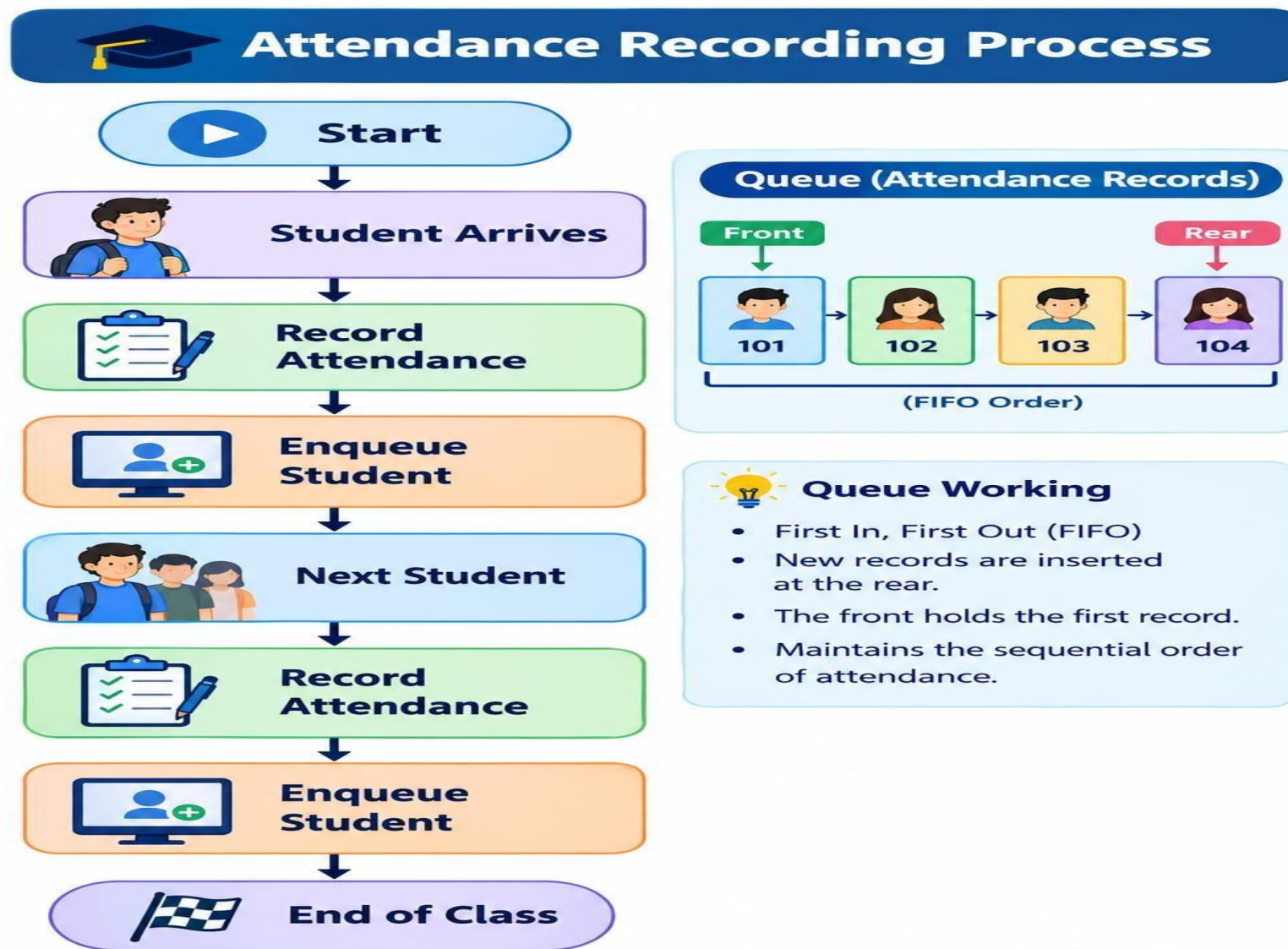


Advantages of Queue

- Maintains student attendance sequence.
- Supports efficient insertion.
- Follows FIFO order.
- Simple to implement.
- New records are added at the rear.
- Suitable for continuous attendance recording.
- Helps process attendance in order.



ATTENDANCE RECORDING PROCESS





Conclusion

- A Queue is suitable for the university attendance system.
- Attendance is recorded sequentially.
- Queue maintains the FIFO order.
- New attendance is inserted at the Rear.
- Enqueue provides $O(1)$ insertion efficiency.
- Queue is simple and efficient for sequential attendance recording.



THANK
YOU